

VENDOR PERFORMANCE ANALYSIS

A PROJECT REPORT

SUBMITTED IN PARTIAL FULFILMENT FOR THE

AWARD OF THE DEGREE

OF

BACHELOR OF COMPUTER APPLICATION



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JAN-MAY 2026

CANDIDATE’S DECLARATION

I hereby certify that the project entitled “ **Vendor Performance Analysis** ” submitted by **Abhay Kumar Upadhyay (2300207)** in partial fulfilment of the requirement for the award of degree of the **BCA** (Computer Science & Engineering) submitted in I.K. Gujral Punjab Technical University, Kapurthala at Baba Banda Singh Bahadur Engineering College, Fatehgarh Sahib is an authentic record of my own work carried out during a period from January, 2026 to May, 2026/ under the guidance of **Dr. Jaspreet Kaur** (Department of Computer Science & Engineering). The matter presented in this project has not formed the basis for the award of another degree, diploma, fellowship or any other similar titles.

Signature of the Student

Place:

Date:



BABA BANDA SINGH BAHADUR ENGINEERING COLLEGE

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(Courses Accredited by NBA (AICTE))



Dr. Lakhvir Singh , Principal

Ref. No. –

Date

CERTIFICATE

This is to certify that the project titled “ **Vendor Performance Analysis** ” is the bona fide work carried out by **Abhay Kumar Upadhyay (2300207)** in partial fulfilment of the requirement for the award of degree of the B. Tech. (Computer Science & Engineering) submitted in I.K. Gujral Punjab Technical University, Kapurthala at Baba Banda Singh Bahadur Engineering College, Fatehgarh Sahib is an authentic record his/her work carried out during a period from January, 2026 to May, 2026/ July, 2026 to December, 2026 under the guidance of Dr. Jaspreet Kaur , Department of Computer Science & Engineering).

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ABSTRACT

The **Vendor Performance Analysis** project is designed to improve the efficiency, transparency, and evaluation of vendor-related data within an organization. This system provides an interactive platform for monitoring, analyzing, and visualizing vendor performance based on key business metrics such as sales, purchase volume, profit margins, and delivery efficiency.

The **User Panel** allows business analysts and managers to access detailed insights into vendor performance, compare different vendors, and identify top-performing as well as underperforming vendors. Users can filter data based on time period, product categories, or vendor details, and generate reports to support data-driven decision-making.

The **Admin Panel** acts as a centralized system for managing vendor data, updating datasets, and maintaining the accuracy of performance metrics. Administrators can monitor overall system performance, manage data integration, and ensure consistency across all vendor-related information.

By integrating data analytics and visualization tools, the **Vendor Performance Analysis** system helps organizations optimize vendor selection, improve supply chain efficiency, and make informed strategic decisions through a clear and user-friendly dashboard interface.

Marks to be filled by Guide	Marks Obtained
Regularity ()	
Self-Motivation and Determination ()	
Working within Team ()	
Total ()	
Signature of the Guide	

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CHAPTER 1

INTRODUCTION

1.1 Brief Overview of Work

The **Vendor Performance Analysis** project is designed to analyze and evaluate vendor performance using data analytics and visualization tools such as Python, SQL, and Power BI. In today's business environment, organizations depend heavily on vendors for procurement and supply chain operations. Therefore, analyzing vendor performance is essential for improving efficiency, reducing costs, and ensuring quality service.

This project demonstrates how data collected from various sources can be processed using Python libraries and SQL queries to extract meaningful insights. These insights are then visualized through an interactive Power BI dashboard. The system helps in tracking key metrics such as sales, purchase quantity, profit margins, and vendor contribution over different time periods.

By transforming raw vendor data into structured and meaningful information, the dashboard provides a clear view of vendor performance. It helps stakeholders identify top-performing vendors, detect underperforming ones, and make informed decisions to optimize business operations and vendor relationships.

1.2 Objectives

The main objectives of the **Vendor Performance Analysis** project are:

1. To analyze vendor performance based on sales, purchases, and profit metrics.
2. To identify top-performing and underperforming vendors.
3. To provide visual insights using interactive dashboards and charts.

1.3 Scope

The scope of the Vendor Performance Analysis project is to analyze and evaluate vendor performance using data analytics and visualization techniques. It focuses on key parameters such as sales, purchase, profit, and vendor contribution. By

using tools like Python, SQL, and Power BI, the project converts raw data into meaningful insights that support effective business decision-making.

This project can be used by organizations to identify high-performing and low-performing vendors, compare performance across different parameters, and monitor trends over time. The interactive dashboard provides a simple and efficient way to understand vendor behaviour and improve operational efficiency.

In the future, the scope of this project can be extended by integrating real-time data, applying predictive analytics, and including more detailed performance metrics. This will enhance its usability for strategic planning, vendor selection, and long-term business growth.

1.4 Project Dashboard

The **Vendor Performance Analysis Dashboard** visualizes vendor-related data across different time periods and categories.

It includes the following components:

- Total Sales and Purchase Overview
- Vendor-wise Performance Analysis
- Profit and Margin Analysis
- Top and Bottom Vendors based on performance
- Sales and Purchase Trends over time (Line Charts)

The dashboard allows users to interact with the data using filters such as vendor name, time period, and product category, making the analysis more flexible and insightful.

1.5 Project Requirements

1.5.1 Hardware Platform

- Processor: Intel i5 or above
- Processor Speed: 1.30 GHz or higher

- RAM: 16 GB or more
- Hard Disk: 1 TB
- Internet Connection: Required
- Display: 1024 × 768 resolution or higher
- Mouse and Keyboard: Standard input devices

1.5.2 Software Platform

- **Python (Pandas, NumPy, Matplotlib, Seaborn)** – for data cleaning, processing, and analysis
- **SQL** – for data querying and extraction
- **Power BI Desktop** – for data modeling, visualization, and dashboard creation.
- **Microsoft Excel** – for data cleaning and preprocessing.
- **Dataset Source** – Vendor sales and purchase dataset (CSV/Excel file)

CHAPTER 2

LITERATURE SURVEY

Literature Survey

Vendor performance evaluation is an important part of supply chain management because it directly affects cost, quality, and delivery efficiency. Over the years, many researchers have studied different methods to evaluate and select vendors based on multiple criteria.

Early work by **Dickson (1966)** identified key factors such as quality, delivery, and past performance as the most important criteria for vendor selection. This study showed that vendor evaluation should not be based only on cost but should include multiple performance factors.

Later, **Weber et al. (1991)** reviewed different vendor selection methods and highlighted the importance of using multi-criteria decision-making approaches. Their study explained that evaluating vendors using only one factor is not effective, and a combination of factors such as cost, quality, and delivery should be considered.

Ellram (1990) introduced the concept of strategic partnerships in vendor selection. According to this study, organizations should focus on building long-term relationships with vendors instead of making decisions based only on short-term benefits. This approach improves trust and overall performance.

Degraeve and Roodhooft (1999) proposed the Total Cost of Ownership (TCO) approach, which considers both direct and indirect costs related to vendors. This method helps organizations understand the hidden costs involved in procurement and make better decisions.

Talluri and Narasimhan (2004) applied Data Envelopment Analysis (DEA) to measure vendor efficiency. Their work showed that mathematical models can be used to compare vendor performance objectively using multiple performance indicators.

In recent years, research has focused more on the use of data analytics and business intelligence tools. **Kumar and Singh (2020)** highlighted that modern analytics-based systems and dashboards help organizations continuously monitor vendor performance and support data-driven decision-making.

Despite these advancements, most studies focus on theoretical models, and there is limited practical implementation using real datasets and modern tools. This project addresses that gap by applying Python, SQL, and Power BI to analyze vendor performance and generate meaningful insights through an interactive dashboard.

2.1 Related background

Vendor performance evaluation plays a critical role in supply chain management as it directly impacts cost efficiency, product quality, and delivery performance. Early studies identified key evaluation criteria such as quality, delivery reliability, and cost efficiency as fundamental factors in vendor selection.

Various research works emphasize the importance of multi-criteria decision-making approaches, where vendors are evaluated based on multiple performance indicators rather than a single factor. This approach provides a more balanced and realistic assessment of vendor capabilities.

Modern research highlights the shift from traditional evaluation methods to analytics driven approaches. Business Intelligence tools and data analytics techniques are now widely used to monitor vendor performance continuously. These tools help organizations analyze large datasets, identify patterns, and make strategic decisions.

Proposed System: Despite these advancements, practical implementation using real datasets and modern tools like Python, SQL, and Power BI is still limited in academic projects. This project addresses this gap by applying data analytics techniques to evaluate vendor performance using structured datasets and interactive dashboards.

2.2 Feasibility Study

A feasibility study is conducted to determine whether the system can be developed efficiently, economically, and operationally to meet the project requirements.

2.2.1 Technical Feasibility

The **Vendor Performance Analysis** system is technically feasible due to the availability of modern tools such as Python, SQL, and Power BI. Python libraries like Pandas and NumPy enable efficient data cleaning and processing, while SQL helps in querying and managing structured data.

The system ensures:

- Efficient handling of large datasets
- Compatibility with data formats such as CSV and Excel
- Interactive and responsive dashboard visualization
- Accurate and fast data processing.

With these capabilities, the project meets all technical requirements and ensures optimal performance for global-scale data analysis.

2.2.2 Economic Feasibility

The project is economically feasible as it uses cost-effective and widely available tools. Python and SQL are open-source technologies, and Power BI Desktop is freely available for development purposes.

Key points include:

- Minimal development cost
- Reduced manual effort and time
- Improved decision-making efficiency
- High return on investment through better vendor management

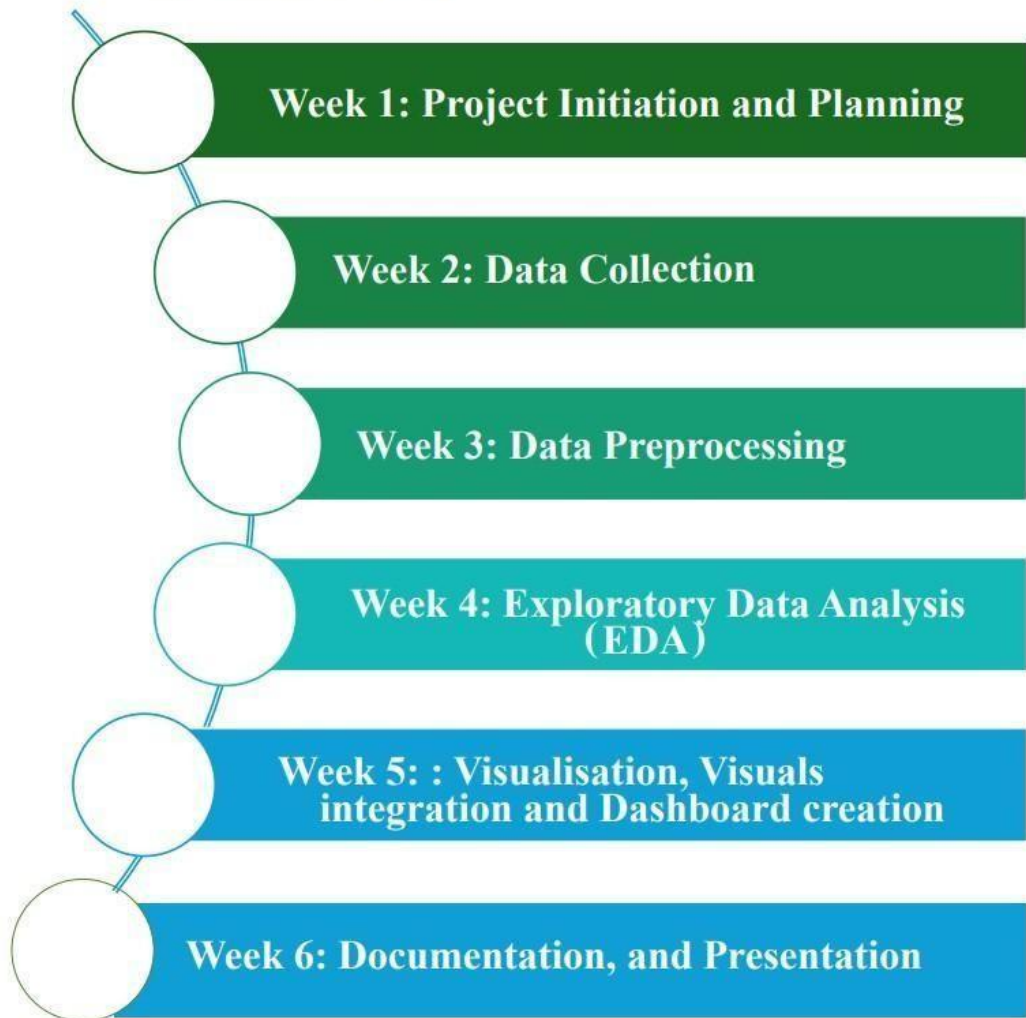
2.2.3. Operational Feasibility

The system is operationally feasible as it is designed to be simple, user-friendly, and easy to use for both technical and non-technical users.

The transition from traditional manual vendor evaluation methods to an automated analytics-based system improves efficiency and accuracy. Users can easily interact with dashboards, apply filters, and generate insights without requiring advanced technical knowledge.

The system enhances organizational performance by enabling continuous monitoring of vendor activities and supporting timely decision-making.

2.3. Project Timeline Chart



CHAPTER 3

SYSTEM ANALYSIS & DESIGN

3.1. Software Tools

Python (Pandas, NumPy, Matplotlib)

Python is a high-level, versatile programming language widely used in data science, analytics, and machine learning. In the **Vendor Performance Analysis** project, Python plays a crucial role in processing raw data and converting it into a structured format suitable for analysis and visualization.

Python provides powerful libraries such as **Pandas** and **NumPy** for efficient data manipulation and numerical operations. Pandas allows handling of large datasets through DataFrames, enabling operations like filtering, grouping, merging, and reshaping data. NumPy supports fast mathematical computations and array-based processing.

For data visualization during exploratory analysis, libraries like **Matplotlib** are used. These libraries help in identifying trends, patterns, and relationships within the vendor dataset before creating the final dashboard.

Python improves the overall efficiency of the system by automating repetitive tasks and ensuring accurate data transformation. It also acts as a bridge between raw data and visualization tools like Power BI.

Key Features:

- Efficient handling of large datasets
- Powerful data manipulation using Pandas DataFrames
- Numerical computation using NumPy
- Data visualization using Matplotlib and Seaborn
- Integration capability with other tools like SQL and Power BI
- Automation of data cleaning and preprocessing tasks

Work Done Using Python:

- Imported raw vendor dataset into Python environment

- Cleaned data by removing missing values and duplicates
- Converted data into structured format for analysis
- Performed grouping and aggregation operations
- Conducted exploratory data analysis (EDA)
- Generated initial visualizations to understand trends

SQL (Structured Query Language)

SQL (Structured Query Language) is a standard language used for managing, querying, and manipulating structured data stored in relational databases. In the **Vendor Performance Analysis** project, SQL is used to extract relevant data and perform operations that prepare the dataset for analysis.

SQL enables efficient handling of large volumes of data through queries that can filter, sort, and aggregate information. It is especially useful when working with structured datasets where relationships between different fields need to be maintained.

Using SQL, data can be organized into meaningful formats, making it easier to analyze vendor performance based on different criteria such as sales, profit, and product categories.

Key Features:

- Efficient data retrieval using SELECT queries
- Filtering data using WHERE conditions
- Data aggregation using functions like SUM, COUNT, AVG
- Sorting and grouping data using ORDER BY and GROUP BY
- Ability to handle large datasets efficiently • Structured and organized data management

Work Done Using SQL:

- Extracted relevant vendor and product data
- Filtered records based on specific conditions

- Performed aggregation for total sales, profit, and quantity
- Organized data into structured format
- Prepared dataset for further processing in Python and Power BI

Power BI Desktop

Power BI Desktop is a business intelligence and data visualization tool developed by Microsoft. It is used to transform processed data into interactive dashboards and reports. In the **Vendor Performance Analysis** project, Power BI is used as the final stage to present insights in a visually appealing and easy-to-understand format.

Power BI allows users to create various types of charts, graphs, and KPIs that help in analyzing vendor performance. It also provides features like slicers and filters, which enable users to interact with the data dynamically.

The tool supports data modeling, where relationships between different tables can be established. It also uses DAX (Data Analysis Expressions) to create calculated fields and measures for advanced analysis.

Key Features:

- Interactive dashboards with filters and slicers
- Data modeling and relationship management
- Advanced calculations using DAX
- Integration with multiple data sources
- Real-time data updates and refresh options
- User-friendly interface for easy visualization

Work Done Using Power BI:

- Imported cleaned dataset into Power BI
- Created data model and relationships
- Designed interactive dashboard
- Visualized vendor performance using charts and KPIs
- Added slicers for filtering data by vendor, time, and category

- Generated insights for decision-making

Microsoft Excel

Microsoft Excel is a widely used spreadsheet application for data entry, organization, and basic analysis. In this project, Excel is used as an initial step for handling raw data before processing it using Python and SQL.

Excel provides a simple and user-friendly interface for viewing datasets, performing basic cleaning, and organizing data into a structured format. It is also useful for quick validation and verification of data.

Key Features:

- Easy data entry and organization
- Basic data cleaning and formatting
- Support for formulas and functions
- Data sorting and filtering
- Compatibility with other tools

Work Done Using Excel:

- Opened and reviewed raw dataset
- Performed initial formatting and cleaning
- Organized data into structured tables
- Verified data before further processing

3.2. SOFTWARE DESIGN

The system design of the **Vendor Performance Analysis** project focuses on structuring the complete workflow of collecting, processing, analyzing, and visualizing vendor-related data using Python, SQL, and Power BI. The design ensures efficient data handling, accurate analysis, and interactive visualization for better decision-making.

1. Data Acquisition

Vendor-related data is collected from sources such as sales records, purchase data, and product details stored in CSV or Excel files. The dataset includes information like vendor name, product category, sales amount, purchase quantity, and profit.

The data may also be imported into SQL databases for structured storage and efficient querying.

2. Data Preprocessing

The collected data is cleaned and transformed using Python libraries such as Pandas and NumPy. This step ensures that the dataset is accurate, consistent, and ready for analysis.

Missing values are handled, duplicate records are removed, and data is formatted properly. Columns such as sales, cost, and profit are converted into appropriate numerical formats.

3. Data Modelling

After preprocessing, the data is structured and modeled for analysis. SQL queries are used to organize and aggregate data, while Power BI is used to create relationships between different data tables.

A structured model is created where vendor data, product details, and time-related data are connected to enable efficient filtering and analysis.

4. Interactive Dashboards

Interactive dashboards are created in Power BI to display vendor performance metrics. These dashboards allow users to explore data dynamically using slicers and filters.

Users can analyze vendor performance based on time period, product category, and other parameters.

5. Visualizations

Different types of charts such as bar charts, line charts, pie charts, and KPI visuals are used to represent vendor data.

Line charts are used to show trends over time, bar charts compare vendor performance, and pie charts display category-wise distribution.

6. Tables and Matrices

Tables and matrices are used to display detailed vendor data and summarized information. These visuals help users compare vendor performance across different categories and time periods in a structured format.

7. KPIs (Key Performance Indicators)

Important KPIs are created to measure vendor performance effectively.

These include:

- Total Sales
- Total Purchase
- Profit and Profit Margin
- Vendor Contribution
- Average Sales per Vendor

These KPIs provide a quick overview of overall performance.

8. Dashboard Design

The dashboard is designed to be simple, user-friendly, and visually appealing. Proper color schemes, layout, and alignment are used to improve readability.

The design ensures that users can easily understand insights without technical knowledge.

9. Reporting and Analysis

Reports are generated using Power BI to analyze vendor performance trends, identify high-performing vendors, and detect areas of improvement.

These reports help in making strategic decisions related to vendor selection, cost optimization, and business growth.

10. Ethical Considerations

All data used in the project is handled responsibly and ethically. Data accuracy and integrity are maintained throughout the process.

Any sensitive information is protected, and the analysis is performed only for academic and analytical purposes.

3.2.1. System Design Flow Chart

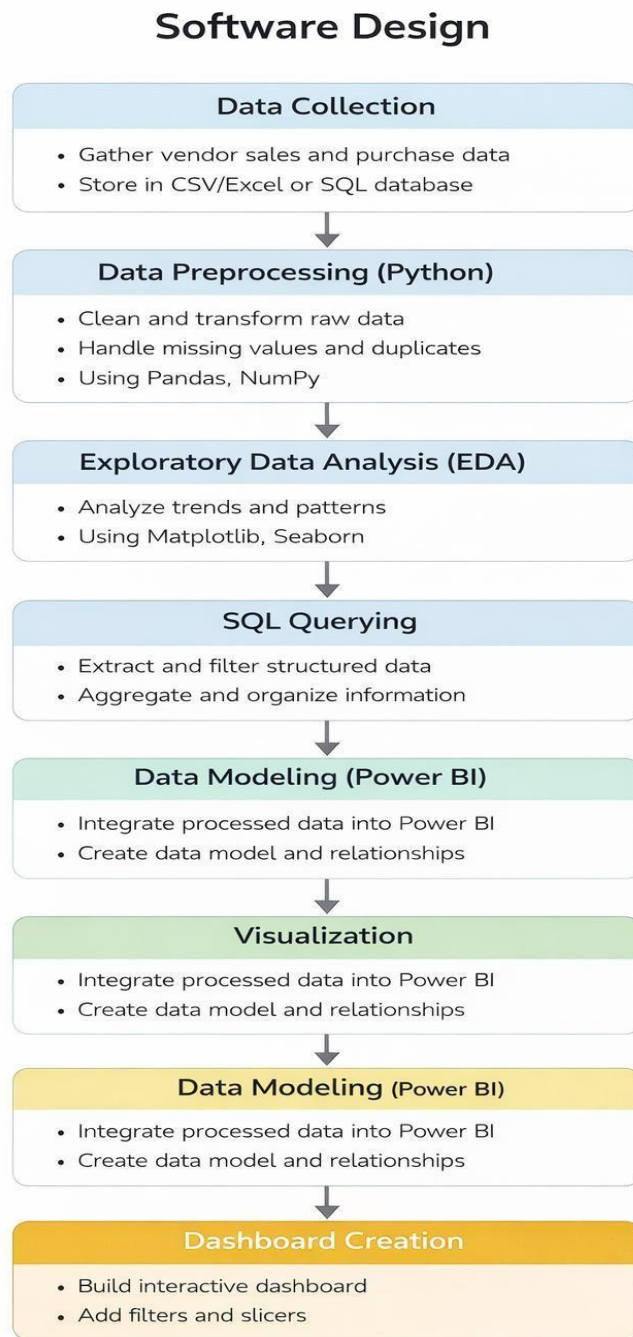


Fig 3.2.1: System Design Flow Chart

3.3. DATA VISUALIZATION

Power BI is a powerful tool for data visualization, allowing you to create interactive and insightful reports and dashboards. Here's a breakdown of key aspects:

1. What is Data Visualization in Power BI

1. **Interactive visuals:** Power BI transforms raw data into engaging visual representations like charts, graphs, maps, and tables.
2. **Insightful storytelling:** These visuals help you uncover patterns, trends, and relationships within your data, making it easier to understand and communicate complex information.
3. **Data-driven decisions:** By presenting data in a clear and concise way, Power BI empowers you to make informed decisions based on evidence.

2. Types of Visualizations in Power BI

Power BI offers a wide variety of built-in visualizations, including:

1. **Basic Charts:** Bar charts, column charts, line charts, pie charts, area charts
2. **Advanced Charts:** Scatter plots, bubble charts, treemaps, heat maps, waterfall charts
3. **Maps:** Filled maps, bubble maps, shape maps
4. **Tables and Matrices:** Simple tables, pivot tables, matrix visuals
5. **Other Visuals:** Gauges, KPIs, cards, slicers

3. Key Features for Effective Visualization

1. **Customization:** Tailor visuals to your specific needs by adjusting colors, labels, axes, and formatting.
2. **Interactivity:** Add interactive elements like filters, drill-downs, and tooltips to allow users to explore data on their own.
3. **Dashboards:** Combine multiple visualizations into a single dashboard to provide a comprehensive overview of key metrics.
4. **Data Connectivity:** Connect to a wide range of data sources, including Excel files, databases, and online services.
5. **AI-powered insights:** Leverage AI features to automatically identify trends, anomalies, and other insights in your data.

4. Best Practices for Data Visualization in Power BI

1. **Choose the right visual:** Select the chart or graph that best suits the type of data you are presenting and the message you want to convey.
2. **Keep it simple:** Avoid cluttering your visuals with too much information. Focus on the key insights you want to highlight.

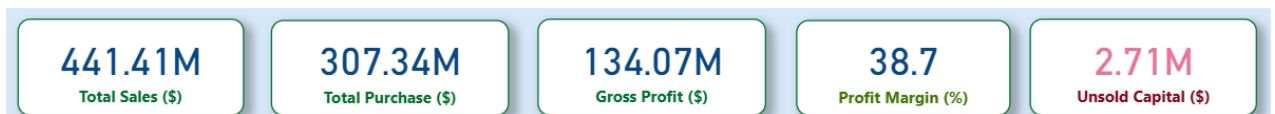
3. **Use clear labels and titles:** Make sure your visuals are easy to understand by using clear and concise labels and titles.
4. **Tell a story:** Use your visuals to tell a compelling story about your data. Guide your audience through the insights you have uncovered.
5. **Test and iterate:** Get feedback on your visualizations and make adjustments as needed to ensure they are clear, effective, and engaging.

3.2.2 Plotting data:

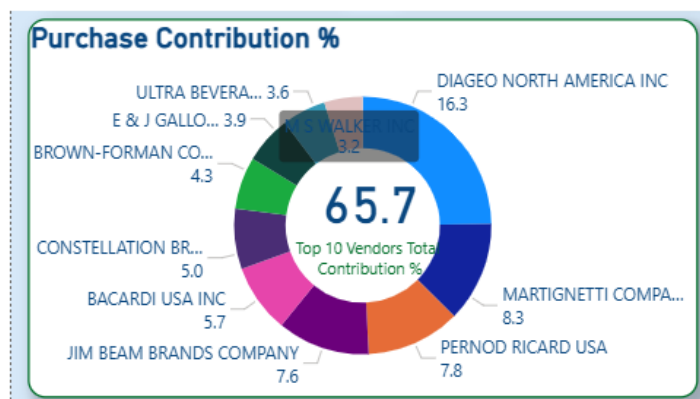
Data is plotted into several types of graphs in Power BI. According to the need, different visualisation will be made.

3.2.3. Data Visualisation

- **Card:** At the top of the dashboard, **card visuals** are used to show important values like total sales, total purchase, gross profit, profit margin, and unsold capital. These cards help users get a quick overview of the overall performance.

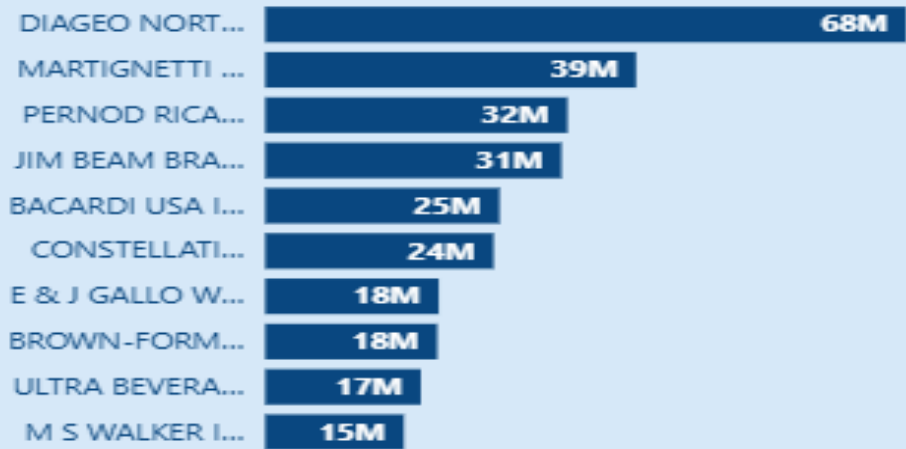


- **Donut chart:** A **donut chart** is used to show the purchase contribution of different vendors. This makes it easy to understand which vendors contribute the most to total purchases.

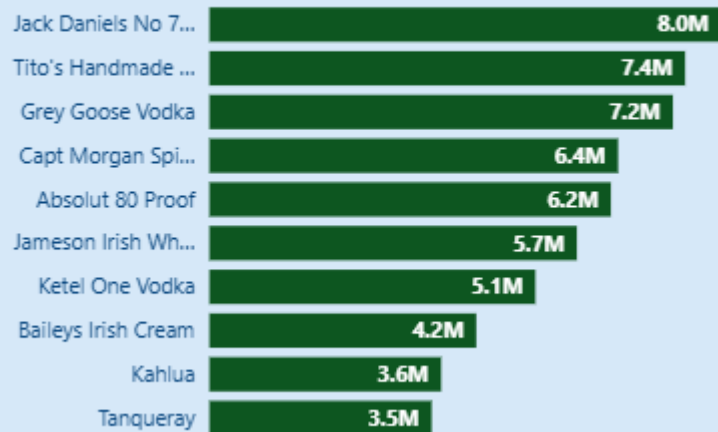


- **Stacked bar chart:** For comparing performance, **bar charts** are used. One bar chart shows the top vendors based on sales, and another shows top brands. These charts help in identifying which vendors or brands are performing well.

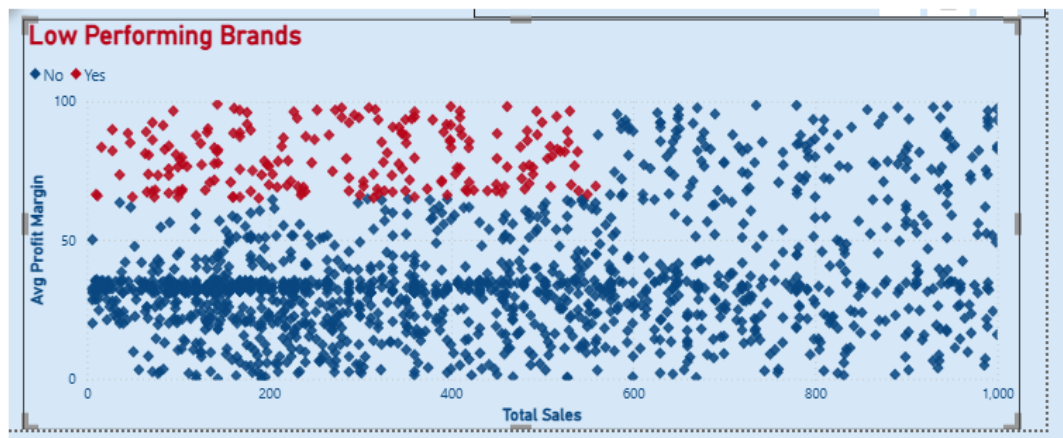
Top Vendors By Sales



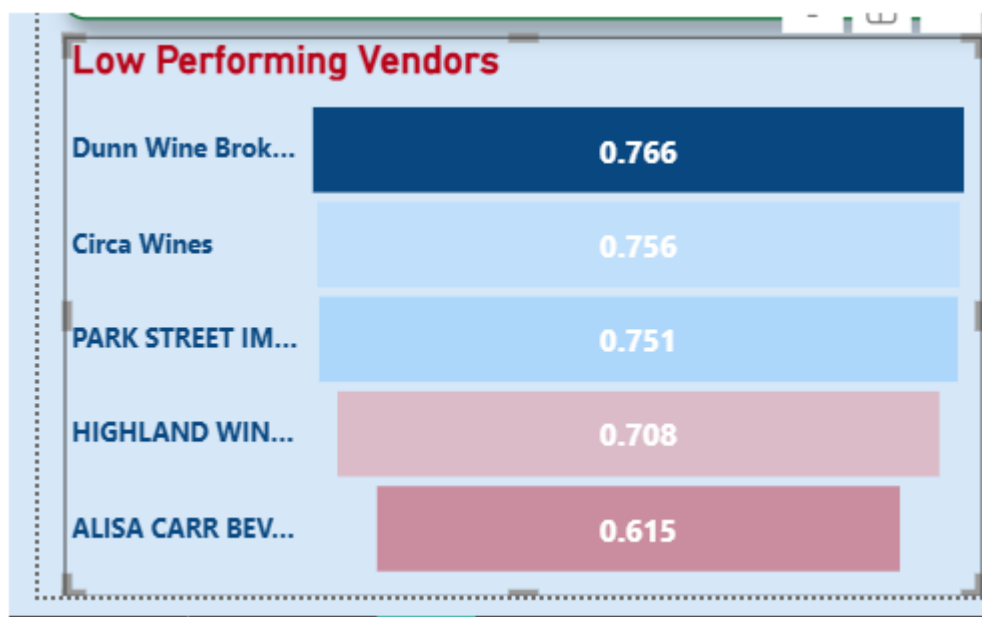
Top Brands By Sales



- **scatter plot:** A **scatter plot** is used to analyze low-performing brands. It helps in understanding the relationship between different values like sales and profit.



- **Funnel :** There is also a bar chart for **low-performing vendors**, where different colors are used to highlight poor performance. This makes it easy to quickly identify problem areas.



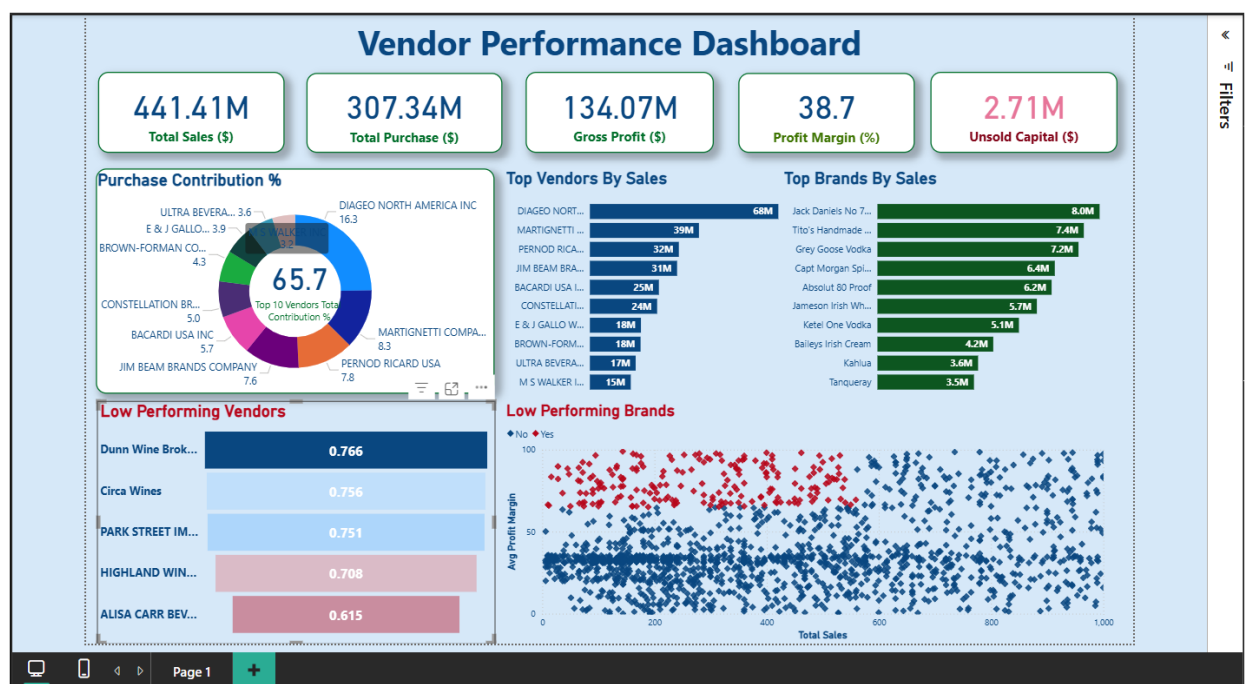
6.3 Creating Dashboards in Power BI

For creating a dashboard in Power BI, various visualisations are combined in a report. Preparing a dashboard in Power BI involves several key steps to transform your data into a visual and interactive dashboard. Here's a high-level

overview of the process:

1. **Connect to Data Sources:** Begin by connecting Power BI to various data sources such as Excel spreadsheets, databases, or cloud services.
2. **Manage Data Model:** Organize your data by creating relationships between different tables and preparing your data model for analysis.
3. **Create Visualizations:** Select from a range of visualizations like charts, graphs, and maps to represent your data.
4. **Arrange Dashboard:** Organize your visualizations logically on the dashboard, grouping related data together.
5. **Enhance Interactivity:** Add features like tooltips, drill-downs, and filters to make your dashboard interactive.
6. **Keep It Simple:** Aim for a clean and uncluttered dashboard design to make it easy to understand.

Initial Dashboard



3.4: IMPLEMENTATION

The **Vendor Performance Analysis** project is implemented using Python, SQL, and Power BI to analyze and visualize vendor performance effectively. The main objective of implementation is to transform raw vendor data into meaningful insights that help in evaluating vendor efficiency and supporting business decisions.

The implementation process begins with collecting raw vendor data from CSV/Excel files. This data is then processed using Python for cleaning and transformation. SQL is used to extract and organize structured data, and finally, Power BI is used to create interactive dashboards and visual reports.

During implementation, various techniques such as data preprocessing, aggregation, and visualization are applied. Python libraries like Pandas and NumPy are used for handling data, while SQL queries help in filtering and structuring the dataset. Power BI features such as DAX, measures, and calculated columns are used to generate key metrics like total sales, profit, and vendor performance indicators.

The final output is an interactive dashboard that allows users to analyze vendor performance dynamically using filters, slicers, and visualizations. This helps in identifying top-performing vendors, monitoring trends, and making data-driven decisions.

3.4.1 How to upload raw data into Power BI

To Create Power BI, visualize dashboard, you need to download Power bi desktop and install. Once you have installed, then we need to follow below process (Microsoft Power BI documentation, 2021):

Steps for Importing from Files (Excel, CSV, Text, JSON):

1. **Open Power BI Desktop:** Launch Power BI Desktop.
2. **Get Data:** On the "Home" ribbon, click "Get Data." A dropdown will appear. Choose the file type (e.g., Excel, Text/CSV, JSON).
3. **Browse:** Navigate to the location of your file and select it.
4. **Navigator Window:** A "Navigator" window will open. Here, you can preview the data, select which tables or sheets to import (if applicable), and perform basic transformations (like renaming columns or changing data types) before loading.

5. Load/Transform Data:

- **Load:** Click "Load" to import the data directly into the Power BI model.
- **Transform Data:** Click "Transform Data" to open the Power Query Editor. This is where you can perform more advanced data cleaning, shaping, and transformation (highly recommended).

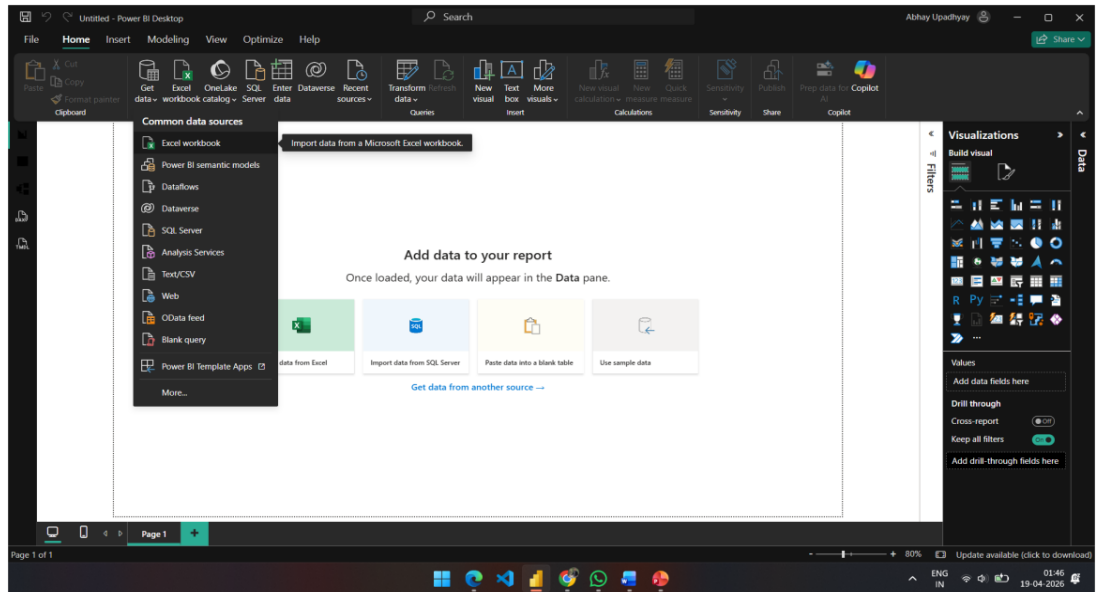


Fig 5.4.1: Upload data into Power BI

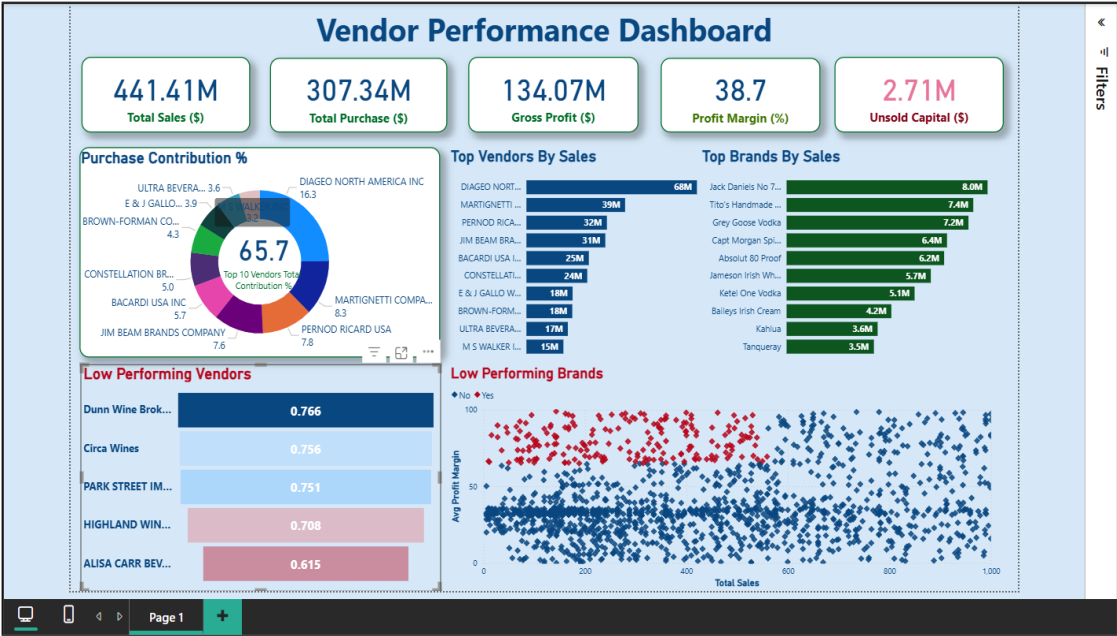
Description	TotalSales	AvgProfitMargin	TargetBrand
Jack Daniels No 7 Black	7964746.76000023	26.5254911201985	No
Tito's Handmade Vodka	7399657.58000011	28.953634467379	No
Absolut 80 Proof	6244752.03000019	27.0519732571601	No
Capt Morgan Spiced Rum	6356320.62000003	27.2617636023475	No
Ketel One Vodka	5070083.56000004	24.2568373268491	No
Grey Goose Vodka	7209908.06000022	25.4138172667723	No
Jameson Irish Whiskey	5715734.69999971	25.189863974263	No
Smirnoff Traveler	2830041.34999993	19.633815114624	No
Tanqueray	3856897.90000008	27.1748266059801	No
Jim Beam	3387851.45999979	32.3008025850949	No
Johnnie Walker Red Label	27110136.8299998	24.7718975247422	No
Dewars White Label	2715618.4299998	24.1928156146114	No
Kendall Jackson Chard V1 RSV	2326007.77999991	37.2283974045845	No
Patron Silver Tequila	2841873.05999977	20.1711633166028	No
Johnnie Walker Black Label	2409124.66999989	24.0120891463688	No
Kallia	3604858.63999953	25.1780728257824	No
Ballies Irish Cream	4150122.06999952	27.7840803298367	No
Crown Royal	3050260.55999962	23.422838285852	No
Capt Morgan Original Barrel	1635666.57999994	19.4554996798638	No
Bombay Sapphire Gin	2668756.41999974	26.9603073019437	No
Stolichnaya	1930270.40999978	25.5427053966346	No
Pinnacle Vodka	2010777.9999972	26.4045142763304	No
Canadian Club	1819338.34999972	28.4035126614486	No
Jose Cuervo Especial	2580342.83999991	28.2248911165714	No
Bacardi Superior Rum	3764116.05999996	36.2158010315046	No
Veuve Clicquot Brut Yellow	1791728.37999991	30.1297720632958	No
Three Olives Vodka	1555364.13999976	47.2668784810409	No

Fig 5.4.2: Load data into Power BI

CHAPTER 4

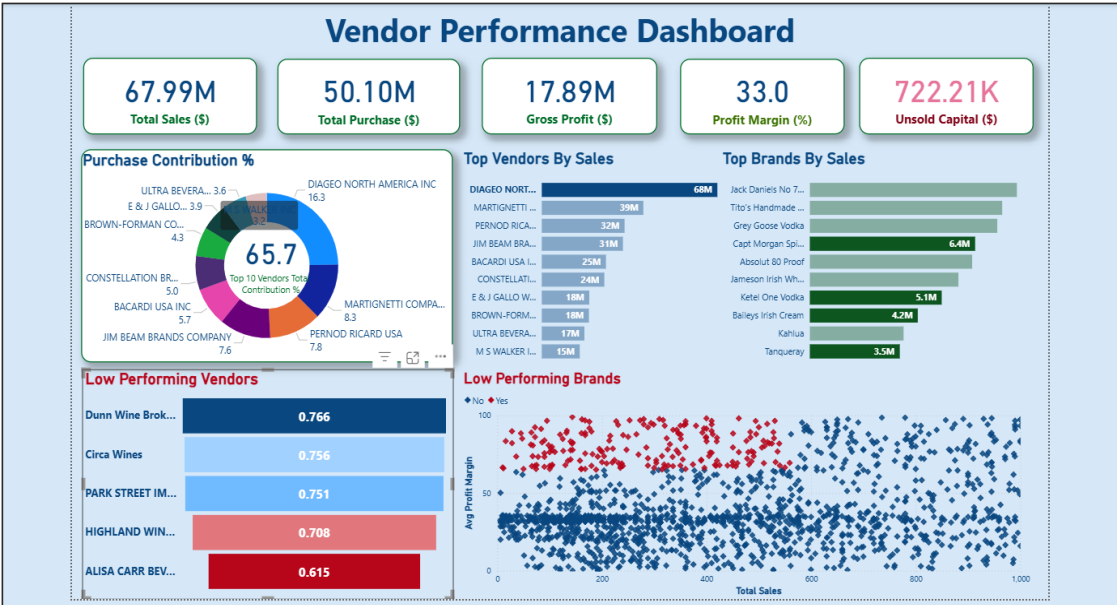
RESULTS

4.1. Data Dashboard



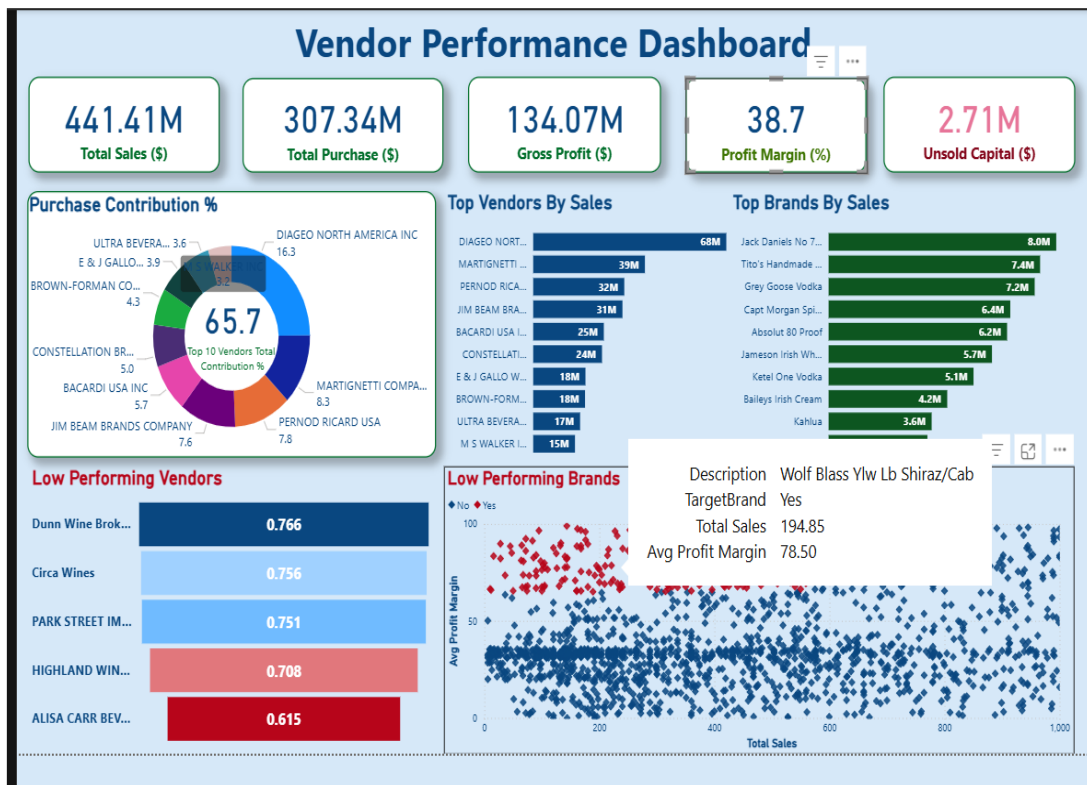
4.1.1 final dashboard

4.2 Data Dashboard depend on DIAGEO NORTH AMERICA INC

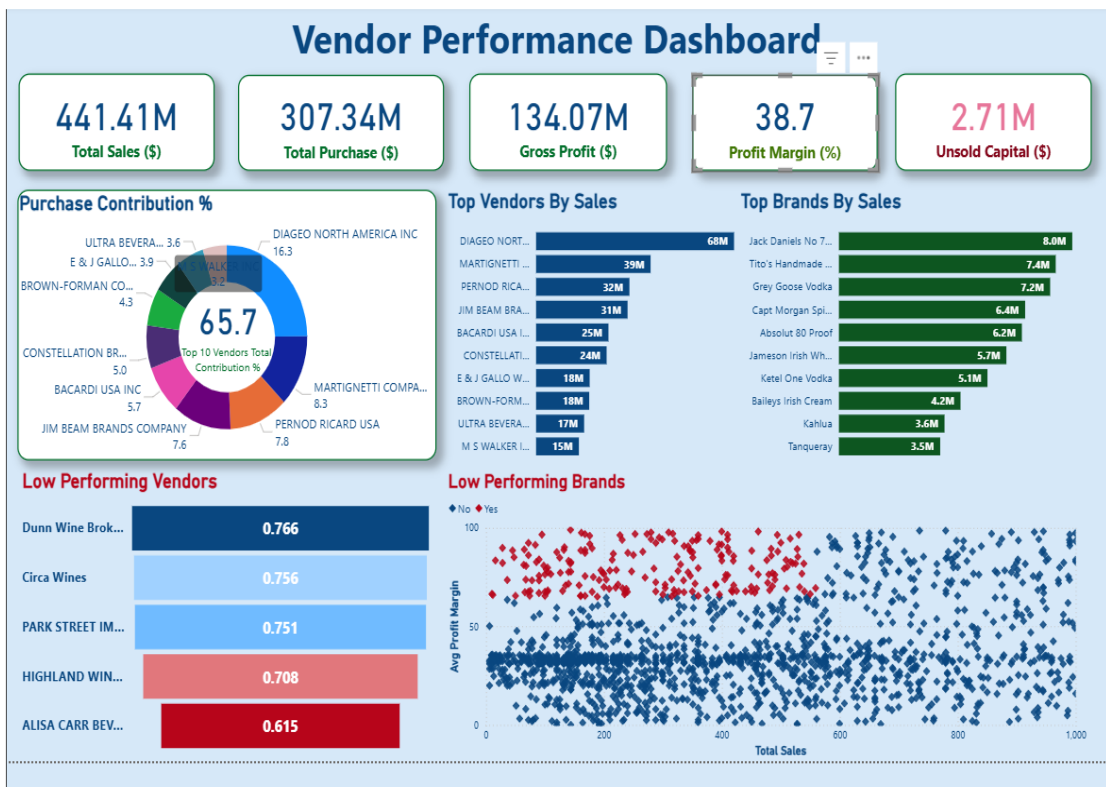


4.4.2 specific vender-based dashboard

Low performance:



Profit margin 38.7%



CHAPTER 5

CONCLUSIONS AND FUTURE SCOPE

5.1 Conclusion

Drawing a definitive conclusion about vendor performance based solely on data visualization is not exhaustive. This project provides descriptive analytics into procurement efficiency, vendor reliability, and profitability trends, but further integration of real-time supply chain data would enhance its decision-making value.

From The Dashboard This Work We Conclude:

- **Operational Efficiency:** The dashboard confirms a near-perfect correlation (\$0.999\$) between purchase and sales, indicating a highly synchronized inventory system where stock is moving as per demand.
- **Vendor Concentration Risk:** A significant insight is that 10 vendors control over 65% of the total business, highlighting a high dependency that could lead to supply chain vulnerabilities.
- **Capital Blockage:** We can identify that \$2.71M in capital is currently locked in unsold inventory, with a major portion concentrated in specific vendors like Diageo North America (\$722K).
- **Profitability Paradox:** High-margin brands (e.g., Crown Royal Apple at 89.8%) are identified as low-volume sellers, suggesting that our current pricing or promotion strategy for premium items needs adjustment.
- **Cost Optimization:** The data validates that bulk purchasing reduces unit costs by up to **72%** (\$10.78 vs \$39.05), proving the economic benefit of large-scale procurement for high-turnover SKUs.

5.2 Future Work

The **future scope** of the **Vendor Performance Analysis** project is to make the system more advanced and useful for real-time business decision-making. Currently, the project works on historical data, but in the future, it can be improved by connecting it to live data sources so that vendor performance can be monitored in real time. Another important improvement can be the use of predictive analytics. By using machine learning techniques, the system can predict future sales, vendor performance, and demand trends. This will help organizations take better decisions in advance instead of only analysing past data.

The project can also be enhanced by adding more detailed features such as vendor rating systems, delivery performance tracking, and customer feedback analysis. This will provide a more complete view of vendor performance and help businesses improve efficiency, reduce costs, and build better vendor relationships.

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